

Article 0.11: Scope and Limits of the Spectral Reduction Lemma – A Pedagogical Synthesis (Revised Edition – 33 Problems)

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Abstract

The Spectral Reduction Lemma (Kithima's Lemma) and its algorithmic completeness theorem provide a powerful method for solving discrete decision problems via ground state extraction. However, not every problem can be encoded to satisfy the four pillars. This article clarifies the scope and limits of the lemma by:

- Summarising the conditions under which a problem is amenable to the four core strategies (CD, EB, FV, FD).
- Listing the **thirty-three problems** already resolved in their definitive order.
- Explaining why advanced strategies (FM, DUST, SFF, Backward Transfer, CTP) remain open, while SRH, SRP, SUTL, SHS are complete.
- Discussing the incomplete spectral approach to Navier-Stokes as an example of a promising but not yet definitive line of research.
- Identifying classes of problems that are fundamentally out of reach (smooth Poincaré in dimension4, Iliev-Sendov, most Erdős conjectures, Cramér, etc.).
- Distinguishing between problems that are potentially approachable via advanced methods (Collatz, Hodge, odd perfect numbers) and those that are fundamentally incompatible.

The goal is to provide a pedagogical map for researchers.

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1 Introduction

The spectral reduction lemma has been successfully applied to **thirty-three major open problems**, including four Millennium Prize Problems ($P = NP$, Yang–Mills mass gap, Riemann hypothesis, Birch–Swinnerton-Dyer), the Fuglede conjecture (dimensions 1–4), the Barnette conjecture, the infinitude of several prime families, and many others. Its power comes from the ability to translate a problem into a finite SAT instance and then into a spectral Hamiltonian whose ground state reveals the solution. However, not every problem fits this paradigm. In this article we systematically describe the domains where the lemma succeeds and those where it fails, with a special emphasis on the conditions required for each of the four core strategies and the obstacles faced by advanced strategies.

2 The four core strategies: conditions for success

The four core strategies (CD, EB, FV, FD) share a common structure:

- A finite configuration space (hypercube or bounded-degree graph).
- A diagonal potential $\hat{V}$ defined by a polynomial-size boolean circuit (primality test, equality test, etc.).
- A constant or polynomial spectral gap (from the expander or hypercube).
- A logarithmic area law (via Brandão–Horodecki and Hilbert curve).

These conditions are satisfied for any problem that can be reduced to SAT (by Cook-Levin) or to a finite search with an effective bound. Consequently, the thirty-three resolved problems all meet these criteria.

3 Advanced strategies: status and conditions

3.1 Strategies that have produced complete proofs (accepted)

- **SRH (Spectral Riemann Hypothesis)** – proves the Riemann hypothesis (SeriesIV). The Hilbert–Pólya operator is constructed explicitly and its self-adjointness is proved; the Hilbert–Pólya conjecture is a corollary (Article IV.7).
- **SRP (Spectrum of the Rational Points)** – proves BSD (SeriesVI).
- **SUTL (Spectral Unitary Translation Groups)** – proves Fuglede in dimensions 1–4 (SeriesXXXI).
- **SHS (Spectrum of Hamiltonian Spanning cycles)** – proves Barnette (SeriesXXXII).

3.2 Strategies still open (conditions not met)

- **FM (Functional Methods)** – targets Hodge, Collatz. Requires explicit error bounds and uniform spectral gap.
- **DUST (Discrete Unified Spectral Theory)** – for Navier-Stokes (incomplete).
- **SFF (Spectral Fingerprint Framework)** – for Hodge. Needs metric-independent fingerprint and constructive discretisation.
- **Backward Transfer Operator** – for Collatz. Missing an effective bound on cycle size.
- **CTP (Circuit Theoretic Proof)** – for odd perfect numbers. Missing an a priori bound or a global contradiction.

3.3 The Hilbert–Pólya conjecture

The Hilbert–Pólya conjecture is now a corollary of the SRH proof (Series IV, Article IV.7). It is no longer an open strategy.

3.4 The Navier-Stokes spectral approach (incomplete)

A spectral approach for the Navier–Stokes existence and regularity problem has been proposed following the Functional Dynamics (FD) strategy. It is presented in a separate series (not numbered among the 33 resolved problems) as a promising but incomplete attempt.

Why it is not yet definitive: The proposed proof relies on several axioms that are not established results:

1. **Bounded relative estimate:** $\|B(u, u)\| \leq \varepsilon \|Au\| + C_\varepsilon \|u\|$ in the Leray space. This is not a standard result; it essentially asserts that the nonlinear term is a small perturbation of the Stokes operator, which is the core difficulty of the Navier–Stokes problem.
2. **Preservation of a uniform spectral gap after perturbation:** Even if the relative bound holds, the Kato–Rellich theorem only guarantees self-adjointness and a gap for a fixed λ . Proving that the gap remains strictly positive uniformly in any truncation or regularisation parameter requires additional estimates that are not provided.
3. **Implication from spectral gap to global regularity:** The statement “a positive spectral gap for $\mathcal{H}_\lambda$ implies that no finite-time blow-up occurs” is not a known theorem. It would constitute a new spectral characterisation of regularity, which has not been proved.

Thus, while the strategy is promising, it does not constitute an unconditional proof. The Navier-Stokes problem remains open.

What would be required for a definitive proof: For the FD strategy to become a complete proof, one would need to:

1. Prove rigorously the bounded relative estimate in the Leray space.
2. Establish that for a sufficiently small λ , the operator $A + \lambda B$ possesses a strictly positive spectral gap that is uniform with respect to any truncation or regularisation parameter.
3. Prove that this positive spectral gap implies the absence of finite-time blow-up (i.e., global regularity).

Until these steps are completed, the Navier-Stokes spectral proof cannot be considered definitive.

4 Domains successfully covered by the lemma

The following table lists the thirty-three resolved problems in their definitive order (see Article 0.9). The strategies used are indicated.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII) – twin primes	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima-Landau (XIV) – fourth Landau problem	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Maximal determinant (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV
22	$1/3$ – $2/3$ conjecture (XXII)	CD
23	Pillai (XXIII)	EB
24	n -conjecture (XXIV)	EB
25	Vizing (XXV)	CD
26	Erdős–Hajnal (XXVI)	EB
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Kummer–Vandiver (XXX)	FD
31	Fuglede (dimensions 1–4) (XXXI)	SUTL
32	Barnette (XXXII)	SHS
33	Infinitude of prime families (XXXIII)	FV

5 What makes a problem amenable to the core strategies?

A problem is amenable if it can be reduced (in polynomial time) to a finite SAT instance (or to a functional operator with a positive spectral gap) whose satisfiability (or eigenvalue positivity) is equivalent to the truth of the problem. This requires:

1. Finite or effectively bounded search space.
2. Binary representation of candidates polynomial in the input size.
3. Polynomial-size boolean circuit distinguishing solutions.

4. (For FV) an asymptotic existence theorem.
5. (For FD) a spectral transfer operator with a positive gap.

6 Domains that are out of reach (with explanations)

6.1 Potentially approachable via open advanced strategies

The following conjectures could be resolved by advanced spectral methods, provided the conditions listed above are met. Their resolution remains open, but the spectral framework offers a roadmap.

- **Collatz conjecture** – missing effective bound on cycle size (Backward Transfer operator).
- **Hodge conjecture** – needs metric-independent spectral fingerprint and constructive discretisation (SFF).
- **Non-existence of odd perfect numbers** – needs a priori bound or global circuit proof (CTP).

6.2 Problems fundamentally incompatible with the spectral framework

The following classes of problems are unlikely to ever be solved by the spectral reduction lemma, because they lack a natural finite SAT encoding or a spectral operator with a positive gap, or they are inherently continuous or undecidable.

- **Smooth Poincaré conjecture in dimension 4** – geometric/topological; no natural operator.
- **Iliev–Sendov conjecture** (zeros of polynomials) – complex analytic; no finite encoding.
- **Most Erdős conjectures** – excluding Erdős–Hajnal (resolved), most lack effective bounds.
- **Borel rigidity conjecture** – aspherical manifolds; not reducible to SAT.
- **Markov homeomorphism problem** – undecidable in dimensions 4.
- **Whitehead conjecture** – undecidable in general.
- **Cramér’s conjecture** – probabilistic, no effective bound.
- **Hadwiger conjecture** – although it is a finite statement about graphs, the space of potential counterexamples is infinite and no effective bound on the size of a possible counterexample is known. The spectral reduction lemma requires reduction to a single finite SAT instance, which is impossible without such a bound. Hence Hadwiger’s conjecture remains out of reach of the spectral method in its current form.

7 Conclusion

The Spectral Reduction Lemma is a powerful tool that covers a wide range of discrete mathematics, including additive, analytic, and algebraic number theory, exponential Diophantine equations, combinatorics, complexity, cryptography, AI, puzzles, linguistics, quantum field theory (Yang–Mills), and graph theory (Barnette). However, it is not a universal panacea: it fails for many continuous problems, undecidable problems, and statements lacking effective bounds or finite reductions. Understanding these limits is essential for directing future research. The Navier-Stokes spectral approach remains incomplete, illustrating the difficulty of applying the method to PDEs. This article provides a pedagogical map that should help researchers quickly assess whether their problem is likely to be solvable by the spectral method.

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